decay at the same time. It decays over a period of time and this decay is basically a random process. It implies that we cannot predict which nucleus is going to decay or rather we can determine like probabilistic basis (like tossing a coin). We can calculate approximately how many nuclei in a sample are decayed over a period of time.

At any instant t, the number of decays per unit time, called rate of decay $\left(\frac{dN}{dt}\right)$ is proportional to the number of nuclei (N) at the same instant.

$$\frac{dN}{dt} \propto N$$

By introducing a proportionality constant, the relation can be written as

$$\frac{dN}{dt} = -\lambda N \qquad (9.32)$$

Here proportionality constant λ is called decay constant which is different for different radioactive sample and the negative sign in the equation implies that N is decreasing with time.

By rewriting the equation (9.32), we get

$$dN = -\lambda N dt \qquad (9.33)$$

Here dN represents the number of nuclei decaying in the time interval dt.

Let us assume that at time $t = 0$ s, the number of nuclei present in the radioactive sample be N_0. By integrating the equation (9.33), we can calculate the number of undecayed nuclei N present at any time t.

From equation (9.33), we get

$$\frac{dN}{N} = -\lambda\, dt \qquad (9.34)$$

$$\int_{N_0}^{N} \frac{dN}{N} = -\int_{0}^{t} \lambda\, dt$$

$$\left[\ln N\right]_{N_0}^{N} = -\lambda t$$

$$\ln\left[\frac{N}{N_0}\right] = -\lambda t$$

Taking exponentials on both sides, we get

$$N = N_0\, e^{-\lambda t} \qquad (9.35)$$

[Note: $e^{\ln x} = e^{y} \Rightarrow x = e^{y}$]

Equation (9.35) is called the law of radioactive decay. Here N denotes the number of undecayed nuclei present at any time t and N_0 denotes the number of nuclei present initially time $t=0$. Note that the number of atoms is decreasing exponentially over the length of time. This implies that the time taken for all the radioactive nuclei to decay will be infinite. Equation (9.35) is plotted in Figure 9.26.

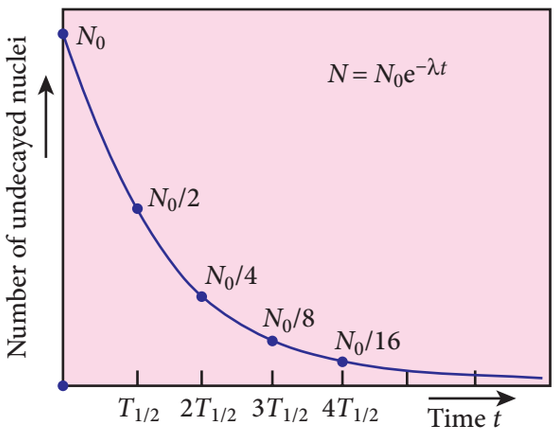


Figure 9.26 Law of radioactive decay

We can also define another useful quantity called activity (R) or decay rate which is the number of nuclei decayed per second and it is denoted as $R = \left|\frac{dN}{dt}\right|$. Note that activity R is a positive quantity.

From equation (9.35), we get

$$R = \left|\frac{dN}{dt}\right| = \lambda N_0\, e^{-\lambda t} \qquad (9.36)$$

$$R = R_0\, e^{-\lambda t} \qquad (9.37)$$

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